1. Design a class Hierachy (pseudocode)

```
CLASS Person
METHOD read Person()
    Read name, age
END METHOD
END CLASS
CLASS STUDENT Inherits Person
    DATA regNo, marks[5], average
    METHOD calculateAverage()
        SUM = 0
        FOR i = 0 TO 4
            SUM ← SUM + marks[i]
        END FOR
            SUM ← sum + Marks [i]
        END FOR
        average ← sum/5
    END METHOD
END CLASS
        METHOD displayDetails()
        Print name
        Print age
        Print regNo
        Print average
        Print
    END METHOD
    END CLASS.
```

2. Complete the missing Pseudocode

```
FOR i ← 0 TO N-1
    FOR j = i+1 TO N-1
        If A[i] == A[j]
            A[j] = -1
        END IF
    END FOR
END FOR
```

3.

RETURN length + length.

4. Trace the Execution

Sum = 0

i = 1 → sum = -1

i = 2 → sum = 1

i = 3 → sum = -2

i = 4 → sum = 2

i = 5 → sum = 3

Output = -3

5. Write efficient pseudocode

```
BEGIN
READ N
READ ARRAY A
largest ← -∞
second ← -∞
FOR i ← 0 TO N-1
 If A[i] > largest THEN
     Second ← largest
     largest ← A[i]
 ELSE IF A[i] > Second AND A[i] ≠ largest THEN
     SECOND = A[i]
     END IF
 END FOR
 PRINT SECOND
 END
```

6. Convert Problem statement into Pseudo code

```
INTERFACE Payment
METHOD payment
END
CLASS CreditCard IMPLEMENTS Payment
CLASS UPI IMPLEMENTS Payment
CLASS NETBanking IMPLEMENTS Payments
```

BEGIN

READ choice

CALL Pay (amount)

END

7. Fill in the blanks

Inherits

New Dog()

8. Find logical Error

CONSTRUCTOR (id, name)

this.id = id

this.name= name

END CONSTRUCTOR

9. Case study

CLASS vehicle

DATA vehicle No.

METHOD display()

END

CLASS car INHERITS vehicle

DATA fuel Type

END

CLASS truck Inherits vehicle

DATA load capacity

END

BEGIN

CREATE objects

STORE objects

Display objects

END